

PRST (20/11/2021)

Ex 3

$$\begin{aligned}
 & P(|Y| < X^3) \\
 &= \int_0^1 P(|Y| < x^3 / X=x) f_X(x) dx \\
 &= \int_0^1 \left(\frac{2x^3}{2x} \right) 2x dx \\
 &= \frac{1}{2}
 \end{aligned}
 \quad \left| \quad \begin{aligned}
 & P(A) \\
 &= \sum_{i=1}^{\infty} P(A \cap B_i) \\
 &= \sum_{i=1}^{\infty} P(A|B_i) P(B_i)
 \end{aligned}
 \right.$$

Ex 4

$X \quad V(X) = 2$
 $b = 2$

$\mu = E(X)$

$P(|X - \mu|^4 \geq 16) \leq$

$\frac{1}{2} = \underline{16}$

$P(|X - \mu| \geq b) \leq \frac{V(X)}{b^2}$

$P(|X - \mu|^4 \geq 16) = P(|X - \mu| \geq 2) \leq \frac{2}{2^2} = \frac{1}{2}$

Ex 5

$F(x, y) = \frac{1}{2} (x^3 y + x y^3), \quad x \in [0, 1]$
 $y \in [0, 1]$

$P(X < \frac{1}{2})$

$G(x) = F(x, 1) = \frac{1}{2} (x^3 + x)$

$P(X < \frac{1}{2}) = G(\frac{1}{2}) = \frac{1}{2} \left(\frac{1}{2^3} + \frac{1}{2} \right)$
 $= \frac{5}{16}$

$F(x, \infty) = G(x)$
 $F(\infty, y) = H(y)$

Ex 5

$f(x) = \lambda e^{-\lambda x}, \quad x > 0$

$E(e^{3x}) = \frac{\lambda}{\lambda - 3}$

$\int_0^{3x} e^{-\lambda t} f(t) dt$

$< x >$

$$E\left(\binom{3x}{e}\right) = \frac{1}{\lambda - 3}$$

$$E\left(\binom{tx}{e}\right) = \frac{\lambda}{\lambda - t}$$

$$E\left(\binom{5x}{e}\right) = \frac{1}{\lambda - 5}$$

ξ_x $x_1, \dots, x_n$ σ^2

$$m = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\mu = \sqrt{m} \left(1 - \frac{\sigma^2}{8m^2} \right)$$

$$\mu = \frac{1}{n} \sum_{i=1}^n \sqrt{x_i}$$

$$= \frac{1}{n} \sum_{i=1}^n \left(x_i \right)^{\frac{1}{2}}$$

$$= \frac{1}{n} \sum_{i=1}^n \left(y_i + m \right)^{\frac{1}{2}}$$

$$= \frac{1}{n} \sum_{i=1}^n \left[m \left(1 + \frac{y_i}{m} \right)^{\frac{1}{2}} \right]$$

$$= \sqrt{m} \left[\frac{1}{n} \sum_{i=1}^n \left(1 + \frac{y_i}{m} \right)^{\frac{1}{2}} \right]$$

$$= \sqrt{m} \left[\frac{1}{n} \sum_{i=1}^n \left(1 + \frac{1}{2} \frac{y_i}{m} + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!} \left(\frac{y_i}{m} \right)^2 + \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)}{3!} \left(\frac{y_i}{m} \right)^3 + \dots \right) \right]$$

$$= \sqrt{m} \left[1 - \frac{1}{8} \frac{\sigma^2}{m^2} \right]$$

$$\left(\frac{x_i}{m} \right) \leq 1$$

$$\left(\frac{x_i}{m} \right)^p$$

$$y_i = x_i - m$$

$$\text{mean}(y) = 0$$

$$\text{sd}(y) = \sigma$$